The Hidden Cost of Inaction

How EU Carbon Border Fees Create a Strategic Case for Saudi Carbon Pricing

A policy analysis combining KAPSARC's KGEMM macroeconomic model with a novel CBAM trade exposure module to quantify the net economic cost of domestic carbon pricing for Saudi Arabia.

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Bottom line: Without a domestic carbon price, Saudi Arabia pays $0.3–1.5 billion per year in CBAM fees to the EU with zero domestic benefit. Introducing a $30/ton carbon price would recover 3–7% of the economic cost through CBAM savings — and that gap grows as CBAM expands to petrochemicals, the backbone of Vision 2030.

Executive Summary

The European Union's Carbon Border Adjustment Mechanism (CBAM) began imposing fees on carbon-intensive imports in January 2026. Saudi exporters of steel, aluminum, fertilizers, cement, and hydrogen now face a carbon charge at the EU border proportional to their products' embedded emissions. Because Saudi Arabia has no domestic carbon price, its exporters pay the full EU rate — currently $75–85 per ton of CO_2 — with no rebate.

This analysis introduces a dimension missing from existing assessments: the interaction between domestic carbon pricing and CBAM trade exposure. The EU's CBAM includes a rebate mechanism: if an exporting country charges its own carbon price, that amount is subtracted from the CBAM fee. A domestic carbon tax therefore does not just impose an economic cost — it also redirects money from the EU treasury to the Saudi government.

We build a three-layer model. Layer 1 uses published results from KAPSARC's KGEMM to estimate the GDP cost of domestic carbon pricing. Layer 2 calculates sector-level CBAM fees and rebates across seven export sectors. Layer 3 combines them to find the true net economic cost.

> **Key finding: Accounting for CBAM trade effects reduces the estimated cost of a $30/ton Saudi carbon price by 3–7%, worth up to $308 million per year. Under expanded CBAM scope covering petrochemicals and Vision 2030 export growth, the annual CBAM bill without domestic pricing reaches $1.5 billion.**

Recommendation: Saudi Arabia should introduce a sector-targeted carbon price of $20–30 per ton focused on industrial and utility sectors by 2028, ahead of CBAM expansion to petrochemicals. This is not primarily a climate policy — it is a trade competitiveness strategy.

1. Problem Statement

1.1 The CBAM is live and expanding

The EU's CBAM entered its payment phase in January 2026. Importers of steel, iron, aluminum, cement, fertilizers, hydrogen, and electricity into the EU must now purchase CBAM certificates reflecting the embedded carbon emissions of their products. The certificate price tracks the EU Emissions Trading System (ETS) price, currently around €65–80 per ton of CO_2.

The scope is set to expand. The EU Commission has signaled that polymers, organic chemicals, and refined petroleum products will be included by approximately 2030. Full implementation — with zero free allowances for EU producers — is scheduled for 2034. The UK is implementing a parallel mechanism from 2027, and other jurisdictions are considering similar measures.

1.2 Saudi Arabia has maximum exposure

Saudi Arabia has no carbon tax, no emissions trading system, and no domestic carbon pricing mechanism under consideration. This means Saudi exporters face the full CBAM rate with zero rebate. Among major trading economies, Saudi Arabia has one of the highest per-capita emission levels and relies almost entirely on fossil fuels for electricity generation (~98% in 2024). Its industrial emission intensities significantly exceed EU averages across all CBAM-covered sectors.

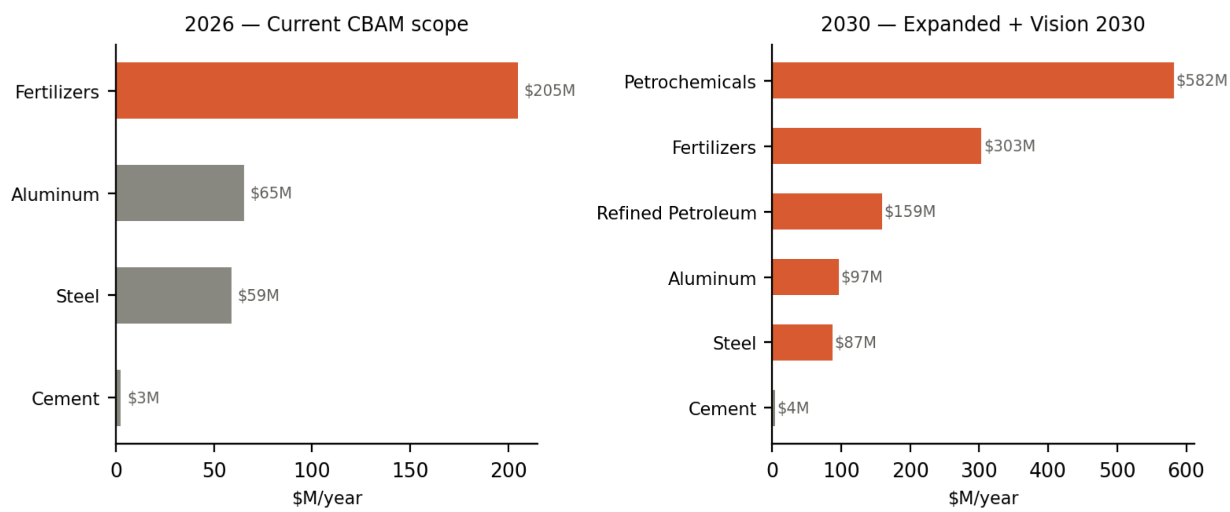

Figure 1. CBAM exposure by sector. Left: 2026 scope (steel, aluminum, fertilizers, cement). Right: 2030 expanded scope including petrochemicals and refined petroleum under Vision 2030 export growth. EU carbon price at $100/ton, no domestic carbon price.

1.3 The Vision 2030 vulnerability

Saudi Arabia's Vision 2030 strategy aims to diversify the economy away from crude oil exports toward higher-value manufactured goods — particularly petrochemicals. SABIC, a wholly-owned subsidiary of Saudi Aramco, is among the world's largest chemical companies and a cornerstone of the diversification strategy. The planned CBAM expansion to polymers and organic chemicals would directly tax the export growth that Vision 2030 depends on.

This creates a paradox: the more successfully Saudi Arabia diversifies into carbon-intensive manufactured exports, the greater its CBAM exposure becomes. Petrochemicals jump from zero CBAM cost today to the single largest exposed sector by 2030, as shown in Figure 1.

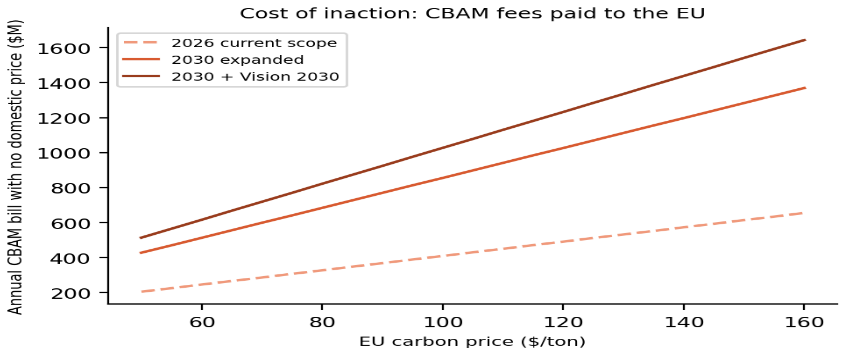

Figure 2. Annual CBAM fees Saudi Arabia pays to the EU with no domestic carbon price, under different CBAM scope assumptions and EU carbon price levels.

2. Methodology

The model has three layers. Layer 1 estimates what happens inside the Saudi economy when a carbon price is introduced. Layer 2 estimates what happens at the EU border. Layer 3 combines them to find the true net cost.

2.1 Layer 1: Domestic effects (KGEMM-derived)

We calibrate domestic emission and GDP response functions from the published results of the KAPSARC Global Energy Macroeconometric Model (KGEMM). Hasanov et al. (2025) report that economy-wide carbon prices of \$20 and \$50 per ton reduce Saudi CO_2 emissions by 5.4% and 9.5% respectively over 2023–2030. They note the response is slightly nonlinear — each additional dollar of carbon tax produces less emission reduction than the last.

We fit a power function to these anchor points:

$$\Delta\mathtt{Emissions}(\tau) = \alpha \times \tau^{\beta}$$

where τ is the carbon price in \$/ton. The calibrated parameters are $\alpha = 0.852$ and $\beta = 0.617$. The value $\beta < 1$ confirms diminishing returns: the cheapest emission reductions (fuel switching) happen first; structural changes to industrial processes are more expensive per ton abated.

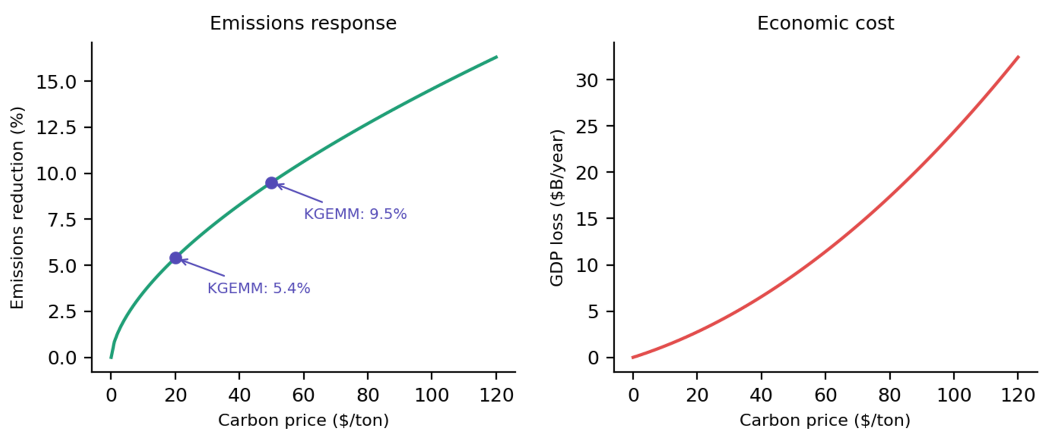

Figure 3. Layer 1 calibration. Left: emission reduction function with KGEMM anchor points (purple dots). Right: GDP loss, modeled as quadratic to capture accelerating costs at higher carbon prices. Source: Hasanov et al. 2025, Energy Economics.

We do not rebuild KGEMM. We treat its published outputs as ground truth for the domestic side. Our contribution is exclusively on the trade side (Layer 2).

2.2 Layer 2: CBAM trade exposure (novel module)

For each CBAM-covered sector, we calculate the fee Saudi exporters pay at the EU border. The core equation:

$$\mathtt{CBAM\_cost}(s) = \mathtt{Exports}(s) \times \mathtt{Excess\_intensity}(s) \times \max(P_{EU} - \tau,\ 0)$$

Where Exports(s) is Saudi export value to the EU in sector s; Excess_intensity(s) is the difference between Saudi and EU emission intensity (tons CO_2 per dollar of export); P_{EU} is the EU ETS carbon price; and τ is the Saudi domestic carbon price. The max(·, 0) term captures the rebate mechanism: Saudi exporters never receive a payment for having a higher domestic price than the EU.

We model seven sectors: steel, aluminum, fertilizers, cement, and hydrogen (covered from 2026), plus petrochemicals and refined petroleum (expected coverage ~2030). Export volumes come from UN Comtrade bilateral trade data. Emission intensities are drawn from IEA sector reports, the World Steel Association, and the International Aluminium Institute.

Data	Source	Access
Saudi emissions by sector	EDGAR / IEA	Free
Saudi exports to EU	UN Comtrade	Free
Sector emission intensities	IEA, World Steel Assoc., IAI	Free
KGEMM carbon pricing results	Hasanov et al. 2025	Published
EU ETS price projections	BloombergNEF, EU Commission	Partial
CBAM regulation	EU Regulation 2023/956	Free

Table 1. Data sources and accessibility.

2.3 Layer 3: Net economic cost

The true economic cost of a domestic carbon price, accounting for CBAM trade effects:

$$\text{True\_cost}(\tau) = \text{GDP\_loss}(\tau) - \text{CBAM\_savings}(\tau)$$

An important framing note: carbon tax revenue is a fiscal transfer from the private sector to the government, not a net economic gain. We report it separately as fiscal information but exclude it from the net cost calculation. The CBAM savings, by contrast, represent real money that stays in Saudi Arabia instead of flowing to the EU — this is a genuine welfare gain.

We sweep τ from \$0 to \$150 per ton across nine scenarios varying three dimensions: EU carbon price (\$60–\$150/ton), CBAM scope (2026 current vs. 2030 expanded), and Saudi export volumes (current vs. Vision 2030 projections).

3. Findings

3.1 The CBAM correction to carbon pricing costs

The central finding: accounting for CBAM trade effects reduces the estimated economic cost of Saudi carbon pricing compared to a domestic-only analysis. The magnitude of the correction depends on CBAM scope, EU carbon price level, and Saudi export composition.

Scenario	GDP Loss ($B/yr)	CBAM Savings	True Cost ($B/yr)	Offset
2026 baseline, EU $81/ton	$4.52	$123M	$4.40	2.7%
2026, EU $120/ton	$4.52	$123M	$4.40	2.7%
2030 expanded, EU $81/ton	$4.52	$257M	$4.26	5.7%
2030 expanded, EU $120/ton	$4.52	$257M	$4.26	5.7%
2030 + V2030, EU $81/ton	$4.52	$308M	$4.21	6.8%
2030 + V2030, EU $120/ton	$4.52	$308M	$4.21	6.8%
2030 + V2030, EU $150/ton	$4.52	$308M	$4.21	6.8%

Table 2. Net economic cost of a $30/ton domestic carbon price across scenarios. GDP loss is from KGEMM-calibrated functions. CBAM savings represent money retained domestically rather than paid to the EU. Offset is CBAM savings as percentage of GDP loss.

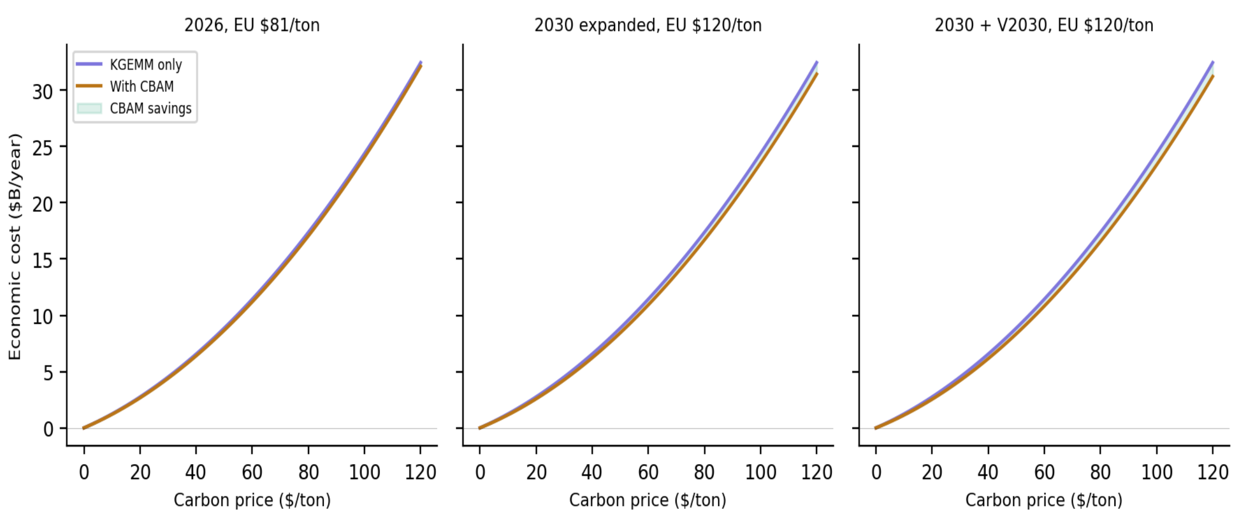

Figure 4. Economic cost of carbon pricing: KGEMM-only estimate (purple) versus CBAM-corrected estimate (amber). The green shaded area represents CBAM savings. The gap widens under expanded CBAM scope and Vision 2030 export growth.

3.2 Sector-level drivers

Under 2026 CBAM scope, fertilizers account for the largest share of Saudi exposure ($253M/year at EU price $100/ton) due to high emission intensity per dollar of export. Aluminum ($81M) and steel ($73M) follow. Cement exposure is minimal because Saudi cement is mostly consumed domestically.

The story changes dramatically under expanded scope. Petrochemicals leap to the largest exposed sector ($485M/year), reflecting both high export volumes to the EU and above-average emission intensity. This is the finding that connects directly to Vision 2030: the sectors Saudi Arabia is betting on for economic diversification are exactly the sectors CBAM will penalize most.

3.3 Sensitivity to EU carbon price

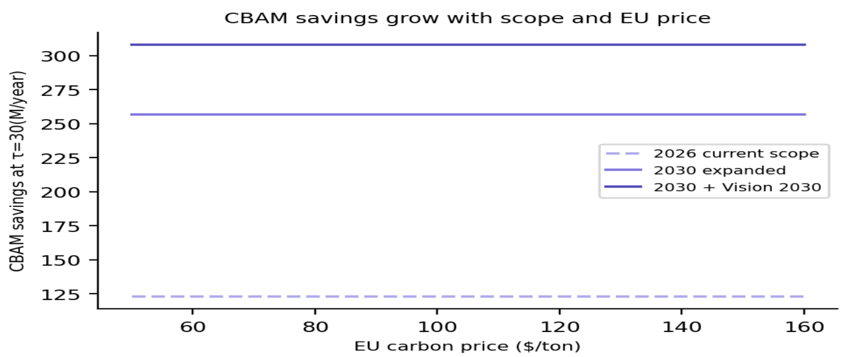

Figure 5. CBAM savings from a $30/ton domestic carbon price at varying EU carbon prices. Savings are larger under expanded scope and higher EU prices, but flatten as the domestic price approaches the EU price (the rebate has a ceiling).

The CBAM savings grow linearly with the EU carbon price up to the point where the domestic price matches the EU price. Beyond that, additional EU price increases do not generate further savings. However, they do increase the cost of inaction — the status quo CBAM bill (Figure 2) rises continuously with the EU price, strengthening the relative case for domestic pricing.

3.4 The cost of inaction

A separate but equally important finding concerns what Saudi Arabia pays under the status quo. With no domestic carbon price (τ = 0), the annual CBAM bill ranges from $331 million (2026, current EU price) to $1.54 billion (2030 expanded scope, EU price $150/ton, Vision 2030 exports). This money flows entirely to EU governments with no offsetting benefit to the Saudi economy — no emission reductions, no domestic revenue, no industrial transition.

> **Under the most aggressive but plausible scenario (expanded CBAM, high EU price, Vision 2030 exports), Saudi Arabia would transfer $1.5 billion per year to the EU — roughly equivalent to the annual budget of several Saudi government agencies — as a pure deadweight cost of policy inaction.**

4. Policy Implications

4.1 Carbon pricing as trade strategy, not climate sacrifice

The conventional framing of carbon pricing in Saudi Arabia treats it as an environmental policy with economic costs. This analysis suggests a different framing: a domestic carbon price is also a trade competitiveness tool. It redirects revenue from EU governments to the Saudi treasury and signals to trading partners that Saudi exports are produced under a recognized carbon pricing regime — which matters as more jurisdictions adopt CBAM-like mechanisms.

This reframing may ease political constraints. A carbon price justified purely on climate grounds faces resistance in an economy where fossil fuels underpin both government revenue and social contracts. A carbon price justified as protecting export competitiveness and retaining fiscal resources has a different political economy.

4.2 Timing relative to CBAM expansion

The window for proactive action is approximately 2026–2029. CBAM's current scope creates moderate exposure ($331M/year). When petrochemicals are added (~2030), exposure roughly triples. Introducing a domestic carbon price before CBAM expansion allows Saudi industry to adjust gradually and demonstrates policy credibility to EU regulators, who have discretion in how they assess foreign carbon pricing regimes.

4.3 Sector targeting and KGEMM's finding

KAPSARC's own analysis shows that pricing carbon only in industrial and utility sectors achieves 74% of the emission reduction of an economy-wide price while generating 72% of the revenue. This is directly relevant: the CBAM only covers industrial sectors, so a sector-targeted domestic price captures most of the CBAM rebate benefit while minimizing the economic disruption of economy-wide pricing. A targeted approach also avoids politically difficult consumer energy price increases.

4.4 Trade-offs and risks

A domestic carbon price has real GDP costs even with the CBAM offset. At $30/ton, the GDP loss is approximately $4.5 billion per year, of which CBAM savings offset only $0.3 billion. The remaining $4.2 billion is borne by industry and consumers through higher energy costs. The government collects roughly $17 billion in carbon revenue, which partially compensates through the fiscal channel but represents a transfer, not a net gain.

There is also a competitiveness risk in non-EU markets. A domestic carbon price raises production costs for all exports, but the CBAM rebate only benefits EU-bound products. For sectors where the EU is a small share of total exports, the net effect of domestic pricing could be negative. Policy design should account for the EU share of each sector's exports.

5. Recommendations

1. **Introduce a sector-targeted carbon price of $20–30/ton by 2028.** Focus on industrial and utility sectors (steel, aluminum, fertilizers, cement, petrochemicals). This captures the CBAM rebate while minimizing consumer impact. The $20–30 range balances meaningful CBAM savings against GDP cost.

2. **Align domestic MRV systems with CBAM reporting requirements now.** CBAM requires verified facility-level emission data. Saudi exporters already face reporting obligations. Building a domestic monitoring, reporting, and verification system serves both CBAM compliance and future domestic pricing. This can begin immediately at low cost.

3. **Earmark carbon revenue for industrial decarbonization.** Use a portion of carbon tax revenue to fund cleaner production in CBAM-exposed sectors. Lower emission intensity directly reduces both domestic tax burden and CBAM fees. This creates a virtuous cycle: the tax funds the investment that reduces the tax.

4. **Engage proactively with EU regulators on CBAM equivalence.** The CBAM regulation allows for recognition of foreign carbon pricing instruments. Early adoption and transparent design increase the likelihood that Saudi Arabia's mechanism receives full credit in CBAM calculations. Delayed or opaque pricing risks partial recognition.

6. Limitations and Further Research

Parameter uncertainty. GDP loss coefficients are calibrated from literature ranges, not directly from KGEMM's internal equations. Access to KGEMM's full model would allow more precise calibration. Sensitivity analysis on the GDP loss function would quantify this uncertainty.

Static sector analysis. The model does not capture dynamic firm responses — Saudi producers investing in cleaner technology in response to the domestic carbon price, which would further reduce CBAM exposure over time. A dynamic extension would likely show larger cumulative CBAM savings.

Scope 2 emissions. CBAM may eventually incorporate indirect emissions from electricity consumption. Given Saudi Arabia's near-total reliance on fossil fuel power generation, this would substantially increase exposure across all sectors.

Trade diversion. We do not model the possibility that Saudi exporters redirect sales from the EU to non-CBAM markets in response to border fees. This is a realistic response in the short term but becomes less viable as more jurisdictions adopt carbon border mechanisms.

Other carbon border mechanisms. The UK CBAM (from 2027) and potential mechanisms in Japan, Canada, and Australia would increase total trade exposure beyond the EU-only estimates presented here.

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